Module 13: How Populations Evolve — Keys to Success

Introduction and Big Picture

Welcome to the mechanics of evolution! Natural selection is not the only way a population can evolve. Evolution simply means a change in a population's genetic makeup over generations. While natural selection is the only mechanism that consistently leads to *adaptation*, other powerful forces—like pure random chance, migration, and physical barriers—can dramatically alter a population's evolutionary path.

Key Information and Concepts

1. The Definition of Microevolution

Microevolution is defined specifically as a **change in allele frequencies** in a population over successive generations. It is evolution on the smallest scale, within a single population.

2. The Five Mechanisms of Microevolution

There are five main forces that can cause allele frequencies to change: 1. **Mutation:** Random changes in DNA sequence. This is the ultimate, originating source of all new genetic variation. 2. **Gene Flow (Migration):** The physical movement of individuals (and the alleles they carry) into or out of a population. (e.g., pollen blowing to a new valley, animals migrating). 3. **Genetic Drift (Chance):** Random fluctuations in allele frequencies. Drift is driven by luck rather than fitness. It is especially powerful in **small populations**, where a random event can drastically alter the gene pool percentage. 4. **Non-random Mating (Sexual Selection):** When individuals preferentially choose mates based on specific traits, shifting the allele frequencies for those preferred traits higher. 5. **Natural Selection:** Environmental pressures favor specific traits that increase survival and reproductive success. This is the only mechanism that consistently leads the population to adapt to its environment.

3. Deep Dive into Genetic Drift

Genetic drift often manifests in two distinct ways: - **The Bottleneck Effect:** A rapid, catastrophic event (like a wildfire, disease outbreak, or flood) indiscriminately kills off a massive portion of the population. The few survivors rebuild the population, but massive genetic diversity is permanently lost. - **The Founder Effect:** A small group of individuals breaks off from a larger population to establish a new colony (like a few birds blown off course to a remote island). The new population's gene pool is entirely limited to the genes of those few original "founders".

4. Three Modes of Natural Selection

When natural selection acts on a measurable trait, it shifts the population's physical traits (phenotypes) in three main patterns: - **Directional Selection:** Shifts the overall makeup of the population by favoring variants at ONE extreme. (e.g., A population of mice gets progressively darker when living on dark volcanic rock). - **Stabilizing Selection:** Removes extreme variants and preserves the intermediate (middle) types. (e.g., Human birth weights—too small is dangerous, too large is dangerous; the average weight has the highest survival rate). - **Disruptive Selection:** Favors variants at BOTH extremes of the distribution, while intermediate types are selected against. (e.g., Finches with large beaks crack tough seeds, small beaks get tiny seeds, but medium-beak finches struggle with both).

5. Hardy-Weinberg Equilibrium

If absolutely none of the five evolutionary forces are acting on a population, it is not evolving. Its allele frequencies stay perfectly stable generation after generation. We call this non-evolving baseline state the **Hardy-Weinberg equilibrium**. True equilibrium requires an infinitely large population, no mutations, random mating, no migration, and no survival advantage for any trait. Because these conditions are virtually impossible simultaneously in the real world, populations are always evolving. However, scientists use the Hardy-Weinberg equations as a mathematical baseline to measure exactly *how fast* and in *what ways* a real population is evolving away from the steady state.

Strategic Tips for Studying

- **Drift vs Flow:** Gene flow relies on *flow* (movement between groups). Genetic drift relies on *drifting* (random luck of the draw).
- **Adaptation:** Always remember that passing a bottleneck doesn't mean an animal was "fit" — it just meant they were lucky. Only natural selection is adaptive.